

# Word Order Variation in Spoken and Written Corpora: A Cross-Linguistic Study of SVO and Alternative Orders

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## Abstract

This study investigates word order variation in spoken and written corpora across five Indo-European languages: English, French, Norwegian (Nynorsk), Slovenian, and Spanish. Using Universal Dependencies treebanks, we analyze the distribution of six canonical word orders (SVO, SOV, VSO, VOS, OSV, OVS). Our results reveal that spoken language consistently exhibits greater word order flexibility than written language. This increased flexibility manifests as a decrease in the dominant SVO pattern and a rise in alternative orders, though the extent of this variation differs across languages. Morphologically rich languages such as Slovenian and Spanish show the most pronounced shifts, while English remains syntactically rigid across modalities. These findings support the claim that modality significantly affects syntactic realizations and highlight the need for typological studies to account for spoken data.

## 1 Introduction

Word order is a fundamental parameter in linguistic typology and syntactic theory. It plays a central role in language classification and shapes our understanding of cross-linguistic variation and syntactic universals. Typological databases such as WALS (Dryer and Haspelmath, 2013) and Grambank (Skirgård et al., 2023) document dominant word order patterns like subject–verb–object (SVO), but these generalizations are typically based on written, formal sources and often fail to capture variation across genres or modalities.

Recent corpus-based work (e.g., Naranjo and Becker, 2018; Levshina, 2019; Baylor et al., 2024) has challenged this categorical view. These studies advocate for a gradient, usage-based approach to word order typology, emphasizing observed token frequencies in syntactically annotated corpora. This shift has enabled a more nuanced classification of languages and has revealed that word order

is not solely a matter of structural constraints, but also reflects contextual factors such as genre, domain, and modality (Levshina, 2019; Baylor et al., 2024).

Despite this growing awareness of contextual variation, there remains a lack of systematic, cross-linguistic studies that focus specifically on modality<sup>1</sup>. While modality is often acknowledged, most existing work incorporates it only indirectly or treats it as a secondary factor within broader genre-based analyses. As a result, cross-linguistic studies that systematically examine the influence of modality on constituent order remain scarce. The extent to which spoken and written language diverge in word order—especially across typologically diverse languages—has yet to be addressed in a comparative framework.

This study offers an exploratory contribution to this gap by examining cross-modal variation in constituent order. We analyze a sample of five Indo-European languages—English, French, Norwegian (Nynorsk), Slovenian, and Spanish—using both spoken and written corpora within the Universal Dependencies framework. Focusing on clause-level syntax, we investigate the distribution of six canonical word orders<sup>2</sup>—SVO, SOV, VSO, VOS, OSV, and OVS—as a typological baseline for comparing modalities. While the tendency for spoken language to show more variation is often assumed, we argue that systematically capturing how this plays out across languages adds empirical weight to such claims and exposes patterns missed in writing-based typologies.

Our goal is to assess whether and how word order differs between speech and writing, and whether these differences follow consistent cross-

<sup>1</sup>Modality here refers to the distinction between spoken and written language.

<sup>2</sup>Canonical refers to the six logically possible SVO permutations; dominant to the most frequent order in a language, based on corpora or typological sources.

linguistic patterns. The analysis is guided by two central research questions: (1) Does word order differ between spoken and written language? and (2) If so, how does it differ?

The remainder of the paper outlines our data and methods (Section 2), presents the results (Section 3), and discusses cross-linguistic trends (Section 4), followed by concluding remarks (Section 5).

## 2 Data and Methods

### 2.1 Corpus Selection and Preparation

The analysis includes five Indo-European languages for which both spoken and written treebanks are available in the Universal Dependencies v2.15 release (Zeman et al., 2024). Selection was limited to spoken-only treebanks, excluding those with mixed genres or code-switching, and required the existence of a corresponding written corpus. Indo-European languages were chosen due to the exploratory scope of the study and the authors’ familiarity with this language group. For English, the *GUM* treebank was used—the largest available resource containing spoken data—which was manually split into spoken and written subsets based on genre metadata.

The *Rhapsodie* (spoken) and *GSD* (written) corpora were used for French (Gerdes et al., 2012; Guillaume et al., 2019); *NynorskLIA* (spoken) and *Nynorsk* (written) for Norwegian (Nynorsk) (Øvrelied et al., 2018; Solberg et al., 2014); *SST* (spoken) and *SSJ* (written) for Slovenian (Dobrovoljc and Nivre, 2016; Dobrovoljc et al., 2017; Dobrovoljc and Ljubešić, 2022); and *COSER* (spoken) and *GSD* (written) for Spanish (Fernández-Ordóñez, 2005–present; Ballesteros et al., 2024).

### 2.2 Data Extraction

We conducted a quantitative analysis using the STARK tool, designed for querying syntactic patterns in UD-formatted dependency trees (Krsnik et al., 2019). For each language and modality pair, we extracted all instances in which a finite verb governs both a nominal subject (nsubj) and a direct object (obj), regardless of clause type—this includes main and subordinate, declarative and interrogative clauses alike. Each sentence was then classified based on the linear order of the subject, verb, and object into one of six canonical word orders: SVO, SOV, VSO, VOS, OSV, or OVS. This procedure yielded a dataset containing word order

distributions for each corpus, which we compared across modalities. To illustrate the six possible orders, Table 1 provides examples in Slovenian with English translations.

With this approach, our analysis aligns with inclusive, usage-based studies (e.g., Gerdes et al., 2019; Östling, 2015; Naranjo and Becker, 2018; Baylor et al., 2024), which aim to capture naturally occurring syntactic variation across clause types. Rather than limiting the analysis to main declarative transitive clauses only (e.g., Levshina, 2019; Dryer, 2013), we include all instances of subject–verb–object structures, regardless of clause type. This enables us to more fully capture modality-sensitive variation, while keeping the analysis straightforward and the results easily interpretable.

| Order | Slovenian                  | Gloss              |
|-------|----------------------------|--------------------|
| SVO   | <i>Mama kupuje jabolka</i> | mother buys apples |
| SOV   | <i>Mama jabolka kupuje</i> | mother apples buys |
| VSO   | <i>Kupuje mama jabolka</i> | buys mother apples |
| VOS   | <i>Kupuje jabolka mama</i> | buys apples mother |
| OSV   | <i>Jabolka mama kupuje</i> | apples mother buys |
| OVS   | <i>Jabolka kupuje mama</i> | apples buys mother |

Table 1: Canonical word order examples in Slovenian. All sentences translate as ‘Mother buys apples’.

## 3 Results

### 3.1 General Observations

Figure 1 summarizes the distribution of six canonical word orders across written and spoken corpora for each language. It shows the relative frequency of SVO, SOV, VSO, VOS, OSV, and OVS, allowing for a direct comparison between modalities.

The results confirm that word order in spoken language differs from written language across all examined languages. In every case, speech exhibits greater variation than writing, with the dominant SVO pattern decreasing in spoken data. Additionally, the degree of flexibility in word order varies across languages, with some showing more pronounced shifts than others. These findings are consistent across the sample.

### 3.2 Language-Specific Findings

**English** shows the least variation. SVO remains dominant, dropping only slightly from 97.4% in written to 93.0% in spoken data. OSV rises modestly from 2.6% to 6.9%, while other patterns remain marginal. This limited shift may suggest that English maintains a relatively high degree of syn-

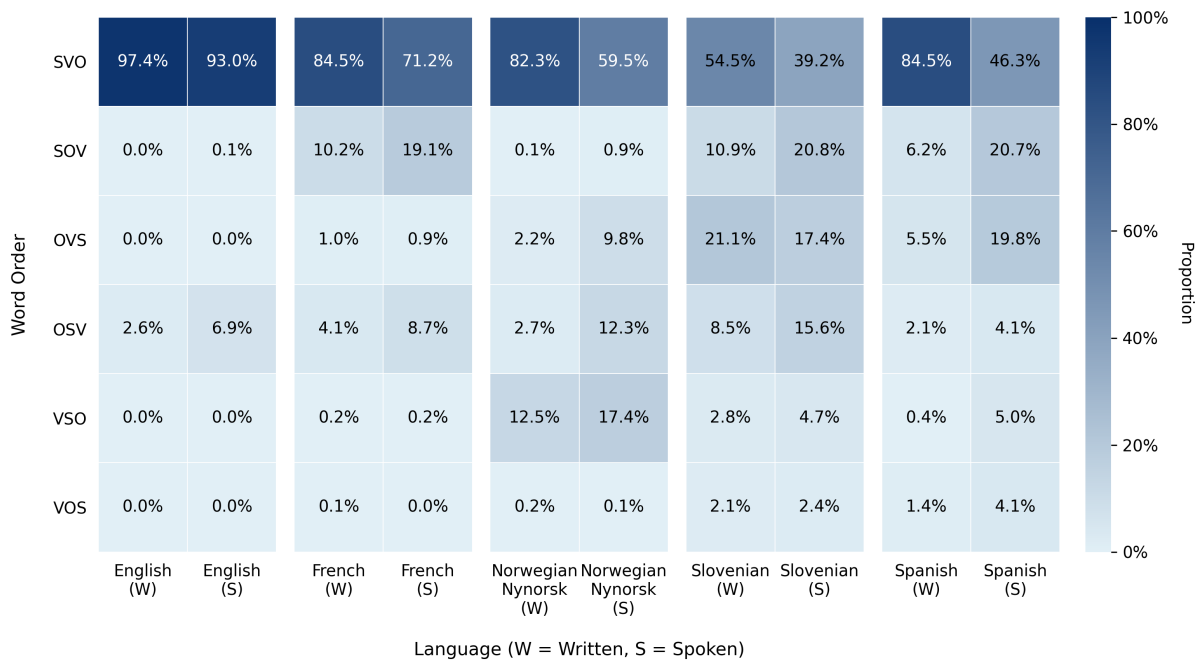


Figure 1: Word order frequencies in written and spoken language across five Indo-European languages.

tactic rigidity even in spontaneous speech. The slight rise in OSV might be due to marked topicalizations or interrogative constructions. For example, *That we did this summer* and *That I give you* illustrate how speakers may foreground the object for emphasis, while questions such as *What did he say?* could also contribute to this pattern.

**French** exhibits a more noticeable change: SVO usage drops from 84.5% to 71.2%, accompanied by increases in SOV (10.2% → 19.1%) and OSV (4.1% → 8.7%). These surface patterns may be influenced by the frequent use of object clitic pronouns and dislocation structures in spontaneous speech. For example, *on l'a mise à l'épreuve* (*we/one her put to the test*) and *je le mets également à l'intérieur* (*I it put also inside*) contain preverbal object clitics, which may lead to an apparent rise in non-SVO orders without indicating a change in underlying syntax.

**Norwegian Nynorsk** displays moderate flexibility. SVO decreases from 82.3% to 59.5%, while OSV and OVS rise significantly (from 2.7% to 12.3% and from 2.2% to 9.8%, respectively). The increase in OSV patterns may partly result from interrogative structures where question elements and pronouns are fronted, as in *Kva den kallast den fóra?* (*What that is called, that feed?*). OVS constructions, such as *Det veit eg ikkje* (*That I don't know*) are common in spoken discourse and may reflect object-fronting for emphasis or information

structure.

**Slovenian** is the most flexible language in the sample. SVO accounts for only 39.2% of spoken clauses, with SOV (20.8%), OSV (15.6%), and OVS (17.4%) forming near-equal shares. This distribution may reflect the language's high degree of pragmatic word order variation. SOV patterns, such as *To mi je šlo zelo na živce* (*That really annoyed me*) and *Jaz ti zdaj pomagam* (*I now help you*) may result from object fronting, emphasis, or prosodic rhythm in spontaneous speech. OSV examples like *To jaz nisem* (*This I am not*) are frequently used for contrastive focus, especially in expressive or corrective contexts.

**Spanish** undergoes the strongest shift from a canonical pattern. SVO falls from 84.5% to 46.3%, while SOV (20.7%), OVS (19.8%), and VSO (5.0%) become more frequent. The increased presence of OV patterns may be partly attributed to clitic constructions and discourse-driven reordering. For instance, *las yuntas lo trillaban* (*the oxen it threshed*) shows preverbal clitic placement that results in an apparent SOV order, while *cómo lo hacía su padre* (*how it did his father*) illustrates an OVS structure that may arise in embedded or emphatic contexts.

## 4 Discussion

Our findings confirm that word order varies significantly between spoken and written modalities

across the examined languages. Although all are classified as SVO-dominant in WALS, spoken data consistently exhibit greater flexibility, with higher frequencies of SOV, OSV, and OVS orders. We observe a cross-linguistic rise in postverbal subjects and object-initial configurations—patterns that are rarely captured in typological descriptions based on written sources.

The extent of this variation differs by language: it is most pronounced in morphologically rich systems such as Slovenian and Spanish, and more limited in structurally rigid languages like English. Morphological richness may allow languages to mark grammatical roles independently of constituent position, supporting more flexible word order. However, this flexibility appears especially prominent in speech, likely due to prosodic, cognitive, and discourse-related factors. Prosody—intonation, stress, and rhythm—may help segment speech and signal syntactic roles, thereby reducing reliance on fixed word order (Levshina et al., 2023). In languages like Slovenian and Spanish, prosodic cues may interact with discourse context to guide constituent placement (Gerken, 1996).

Word order in spoken language can also be shaped by discourse-level considerations, particularly the distinction between given and new information. Prince (Prince, 1981, 1997) as well as Gregory and Michaelis (Gregory and Michaelis, 2001) show that Left-Dislocation in speech promotes new referents to topic status, aligning word order with discourse needs rather than syntactic defaults.

Cognitive and psycholinguistic pressures also seem to shape linearization: speakers tend to place accessible or "given" information early to aid comprehension and allow more time to plan semantically heavier elements (Schouwstra et al., 2022; Levshina et al., 2023). Additionally, spontaneous features of spoken interaction—such as repairs, questions, and topic shifts—may further encourage non-canonical orders, as observed in Slovene dialectal discourse (Kumar, 2019). The fact that the largest spoken shifts occur in structurally flexible systems—such as Slovenian and Spanish—suggests that modality interacts with grammatical properties in meaningful ways. Languages with rich morphological marking provide speakers with the freedom to reorder constituents for pragmatic or processing reasons, especially in spontaneous speech (Levshina, 2019).

These findings highlight the limitations of current typological classifications, which often rely on grammatical descriptions rather than actual usage data—whether written or spoken. Databases like WALS offer valuable generalizations, but their dependence on non-corpus sources means that modality-sensitive variation frequently goes unrecorded. Our results suggest that typologies of constituent order could be refined by incorporating corpus-based evidence. In some cases, languages with a single documented dominant order may in practice display multiple co-dominant patterns, as already recognized in systems labeled “no dominant order.”

Future work should expand this approach to more languages—including those beyond Indo-European—and include additional clause types and genre distinctions. Incorporating entropy-based metrics could further capture the gradient nature of word order variation, complementing categorical classification with usage-based variability. The increasing availability of spoken UD treebanks (e.g., Dobrovoljc, 2022) enables more systematic exploration of this missing dimension, consistent with directions suggested by recent typological research (cf. Baylor et al., 2023).

## 5 Conclusion

This study highlights the significant impact of modality on SVO word order variation across five Indo-European languages. Spoken language consistently shows greater syntactic flexibility, especially in morphologically rich systems like Slovenian and Spanish. These findings challenge typological generalizations based primarily on written data and underscore the need for future studies to incorporate spoken corpora for a more accurate picture of constituent order variation.

## Limitations

This study focuses on five Indo-European languages, limiting typological diversity. Only the Nynorsk variety of Norwegian was included to ensure consistent comparison between spoken and written data.

The corpora vary in size, balance, and genre coverage, particularly between spoken and written modalities, which may influence pattern distribution. Only clauses with overt nominal subjects and objects were included, following WALS criteria, excluding constructions common in morphologi-



cally rich languages where arguments are omitted.

We also restricted the analysis to verbal predicates, excluding nominal and adjectival constructions, and did not distinguish between declarative and interrogative clauses. Our findings are based on quantitative distributions, with no in-depth qualitative analysis.

Lastly, while Universal Dependencies aims for consistency, differences in annotation guidelines or treebank practices may affect comparability. The classification of texts as spoken or written, especially in the *GUM* corpus, is based on metadata, which may not always accurately reflect modality boundaries.

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